

Quantum Mechanics

1] De-Broglie's hypotheses: -

- $\lambda = \frac{h}{mv}$

Where,

λ = wavelength of light

m = mass

v = velocity

h = plank constant

- $\lambda = \frac{h}{p}$

Where,

p = momentum

p = mv

- $\lambda = \frac{h}{\sqrt{2mE}}$

2] Uncertainty principle: -

- $\Delta x \Delta p_x \geq \hbar$

Where,

$\Delta x \rightarrow$ uncertainty in position of particle

$\Delta p_x \rightarrow$ uncertainty in momentum of particle

a. Also $\Delta p_x = m \Delta v_x$

$\Delta v_x \rightarrow$ uncertainty in velocity of particle

- $\Delta E \Delta t \geq \hbar$

Where,

$\Delta E \rightarrow$ uncertainty in energy

$\Delta t \rightarrow$ uncertainty in time spent in excited st

3] Group and Phase Velocity: -

- $V_{\text{group}} = v$

Where,

$v \rightarrow$ velocity of the particle

$v_g / v_{\text{group}} \rightarrow$ velocity of the wave packet or group velocity

- $(V_{\text{group}})(V_{\text{phase}}) = c^2$

Where,

$v \rightarrow$ velocity of the particle

$v_g / v_{\text{group}} \rightarrow$ velocity of the wave packet or group velocity

$v_p / v_{\text{phase}} \rightarrow$ velocity of the wave or phase velocity

4] Schrodinger's equation

- $E_n = \frac{P_n^2}{2m}, \quad E_n = \frac{n^2 h^2}{8mL^2}$

Where,

E_n = energy states of particle

n = no of state

L = width of the area in which particle is enclosed

- $P(x) = \int_0^L [\varphi(x)]^2 dx$

Where,

L = width of the area in which particle is enclosed

$P(x)$ =probability of finding particle within the area

$\Psi_n(x)$ =wave function of particle

- $\Psi_n(x) = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}$

Where,

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